3

Working with MySQL

In this chapter, you'll learn how to connect and log in to MySQL, how to issue MySQL statements, and how to obtain information about databases and tables.

Now that you have a MySQL DBMS and client software to use with it, it would be worthwhile to briefly discuss connecting to the database.

MySQL, like all other client/server DBMSs, requires that you log in to the DBMS before you can issue commands. Your login name might not be the same as your network login name (assuming that you are using a network); MySQL maintains its own list of users internally and associates rights with each user.

When you first installed MySQL, you were probably prompted for an administrative login (often `root`) and a password. If you are using your own local server and are simply experimenting with MySQL, using that login is fine. In the real world, however, the administrative login is closely protected because access to it grants full rights to create tables, drop entire databases, change logins and passwords, and more.

To connect to MySQL, you need the following pieces of information:

- The hostname (the name of the computer), which is `localhost` if you're connecting to a local MySQL server
- The port (if a port other than the default `3306` is used)
- A valid username
- The user password (if required)

As explained in Chapter 2, "Introducing MySQL," all of this information can be passed to the `mysql` command-line utility or entered into the server connection screen in MySQL Workbench.

Note

Using Other Clients If you are using a client other than the ones mentioned here, you still need to provide this information in order to connect to MySQL.

Using the Command-Line Tool

The `mysql` command-line utility is the one client tool that will always be available to you. Even though you are most likely going to use MySQL Workbench as you learn MySQL with this book, it's worth learning how to use the command-line utility, too.

To start using the command-line tool with a locally installed MySQL DBMS, you can use this command:

```
mysql --user=root --password
```

If you are not using the `root` user, specify your username instead. You can specify the password on the command line, or if you just specify `--password`, you'll be prompted to type it after you press Enter.

You then see text that looks something like this:

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 31
Server version: 8.0.31 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

You enter commands at the `mysql>` prompt.

Selecting a Database

When you first connect to MySQL, you do not have any databases open for use. Before you can perform any database operations, you need to select a database. To do this, you use the `USE` keyword.

New Term

Keyword A reserved word that is part of the MySQL language. Never name a table or column using a keyword. Appendix E, "MySQL Reserved Words," lists the MySQL keywords.

For example, to use the `mysql` database, you would enter the following:

► Input

```
USE mysql;
```

► Output

```
Database changed
```

► Analysis

The `USE` statement does not return any results. Depending on the client used, some form of notification might be displayed. For example, the `mysql` command-line utility displays the `Database changed` message shown here upon successful database selection.

Remember that you must always "`USE` a database" before you can access any data in it.

Learning About Databases and Tables

What if you don't know the names of the available databases? And, for that matter, how is MySQL Workbench able to display a list of available databases?

Information about databases, tables, columns, users, privileges, and more is stored in databases and tables themselves. (Yes, MySQL uses MySQL to store this information.) But these internal tables are generally not accessed directly. Instead, you use the MySQL `SHOW` command to display this information (which MySQL extracts from those internal tables). Look at the following example:

▶ Input

```
SHOW DATABASES;
```

▶ Output

```
+--------------------+
| Database           |
+--------------------+
| crashcourse        |
| information_schema |
| mysql              |
| performance_schema |
| sakila             |
| sys                |
| world              |
+--------------------+
7 rows in set (0.00 sec)
```

▶ Analysis

`SHOW DATABASES;` returns a list of available databases. This list might include databases used by MySQL internally (such as `mysql` and `information_schema` in this example). Of course, your own list of databases might not look like the one shown here.

To obtain a list of tables in a database, use `SHOW TABLES;`, as in this example:

▶ Input

```
SHOW TABLES;
```

▶ Output

```
+------------------------------------------------------+
| Tables_in_mysql                                      |
+------------------------------------------------------+
| columns_priv                                         |
| component                                            |
| db                                                   |
```

```
| default_roles                                        |
| engine_cost                                          |
| func                                                 |
| general_log                                          |
| global_grants                                        |
| gtid_executed                                        |
| help_category                                        |
| help_keyword                                         |
| help_relation                                        |
| help_topic                                           |
| innodb_index_stats                                   |
| innodb_table_stats                                   |
| ndb_binlog_index                                     |
| password_history                                     |
| plugin                                               |
| procs_priv                                           |
| proxies_priv                                         |
| replication_asynchronous_connection_failover         |
| replication_asynchronous_connection_failover_managed |
| replication_group_configuration_version              |
| replication_group_member_actions                     |
| role_edges                                           |
| server_cost                                          |
| servers                                              |
| slave_master_info                                    |
| slave_relay_log_info                                 |
| slave_worker_info                                    |
| slow_log                                             |
| tables_priv                                          |
| time_zone                                            |
| time_zone_leap_second                                |
| time_zone_name                                       |
| time_zone_transition                                 |
| time_zone_transition_type                            |
| user                                                 |
+------------------------------------------------------+
38 rows in set (0.00 sec)
+-----------------------+
```

Note

Use of the terms master/slave is ONLY in association with the official terminology used in industry specifications and standards, and in no way diminishes Pearson's commitment to promoting diversity, equity, and inclusion, and challenging, countering and/or combating bias and stereotyping in the global population of the learners we serve.

Analysis

`SHOW TABLES;` returns a list of available tables in the currently selected database. You can also use `SHOW` to display a table's columns:

Input

```
SHOW COLUMNS FROM servers;
```

Output

```
+-------------+-----------+------+-----+---------+-------+
| Field       | Type      | Null | Key | Default | Extra |
+-------------+-----------+------+-----+---------+-------+
| Server_name | char(64)  | NO   | PRI |         |       |
| Host        | char(255) | NO   |     |         |       |
| Db          | char(64)  | NO   |     |         |       |
| Username    | char(64)  | NO   |     |         |       |
| Password    | char(64)  | NO   |     |         |       |
| Port        | int       | NO   |     | 0       |       |
| Socket      | char(64)  | NO   |     |         |       |
| Wrapper     | char(64)  | NO   |     |         |       |
| Owner       | char(64)  | NO   |     |         |       |
+-------------+-----------+------+-----+---------+-------+
9 rows in set (0.00 sec)
```

Tip

The `DESCRIBE` Statement MySQL supports the use of `DESCRIBE` as a shortcut for `SHOW COLUMNS FROM`. In other words, `DESCRIBE customers;` is a shortcut for `SHOW COLUMNS FROM customers;`.

Other `SHOW` statements are supported, too, including these:

- `SHOW STATUS` is used to display extensive server status information.
- `SHOW CREATE DATABASE` and `SHOW CREATE TABLE` are used to display the MySQL statements used to create specified databases or tables, respectively.
- `SHOW GRANTS` is used to display security rights granted to users (all users or a specific user).
- `SHOW ERRORS` and `SHOW WARNINGS` are used to display server error or warning messages.

Note that client applications use these same MySQL commands. Applications that display interactive lists of databases and tables, that allow for the interactive creation and editing of tables, that facilitate data entry and editing, and that allow for user account and rights management all accomplish what they do by using the same MySQL commands that you can execute directly yourself.

> **Tip**
>
> **Learning More About `SHOW`** With the `mysql` command-line utility, you can execute the command `HELP SHOW;` to display a list of allowed `SHOW` statements.

Using MySQL Workbench

The `mysql` command-line utility is an invaluable—and readily available—tool. But it's not exactly intuitive or friendly to use. Fortunately, there is a wonderful interactive tool called MySQL Workbench that we'll be using throughout this book.

> **Tip**
>
> **Obtaining MySQL Workbench** MySQL Workbench is supported on Windows, Mac, and Linux. Refer to Appendix A, "Getting Started with MySQL," for help obtaining MySQL Workbench.

Getting Started

MySQL Workbench needs to know what DBMS you will be using. When MySQL Workbench first runs, it looks for a local DBMS, and if it finds one, it automatically adds the DBMS to the MySQL Connections list, as shown in Figure 3.1.

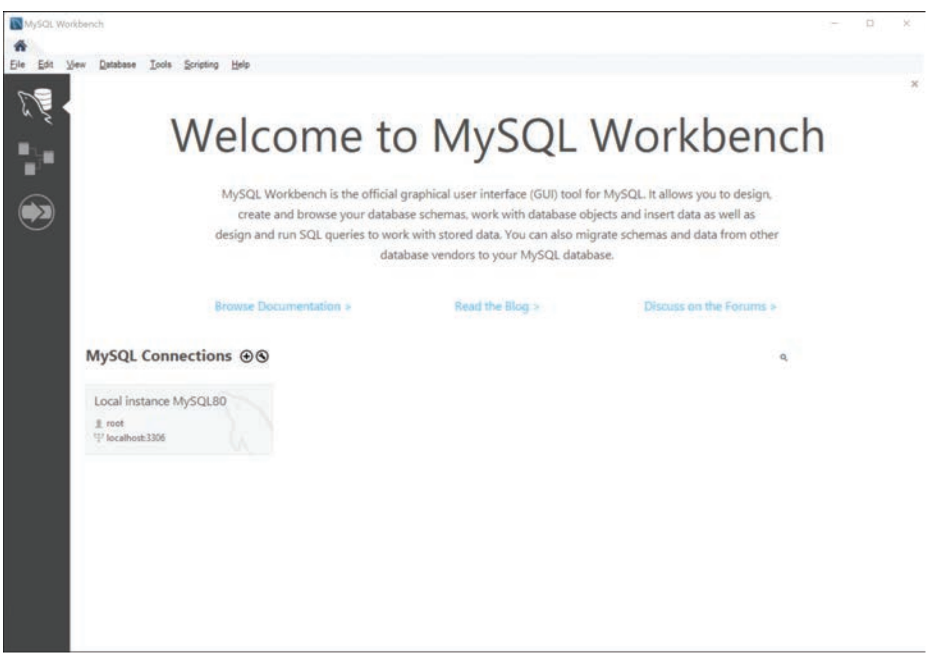

Figure 3.1 DBMS listed under MySQL Connections.

Double-click on the connection, and you are prompted for the password, as shown in Figure 3.2.

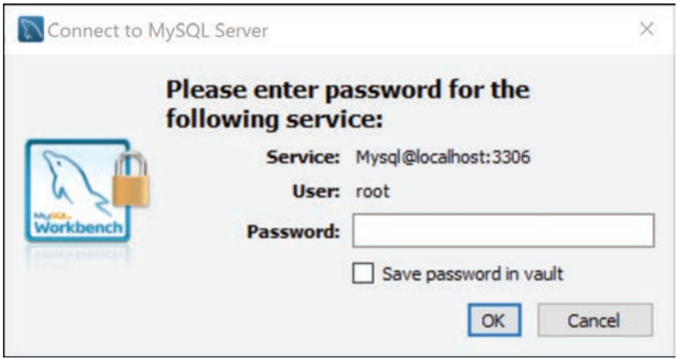

Figure 3.2 The password prompt.

If the login information you enter is correct, you're then connected to the MySQL server.

Using MySQL Workbench

Let's take a few minutes to become familiar with the MySQL Workbench user interface, shown in Figure 3.3.

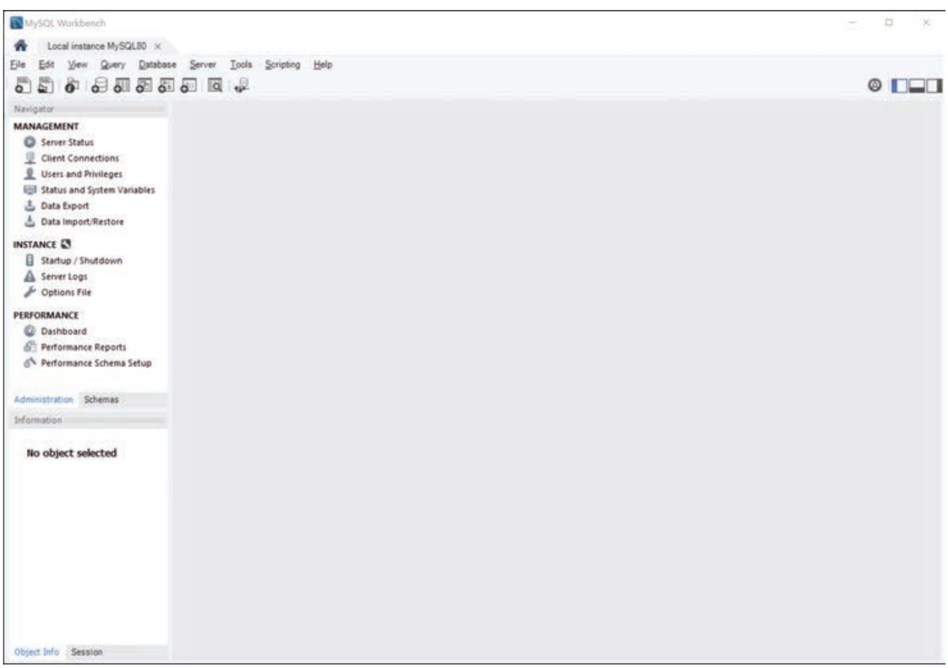

Figure 3.3 The MySQL Workbench user interface.

The upper left is the Navigator. It provides options to manage the server, including displaying server status (much the same as the information displayed previously using the command-line tool).

If you click on the Schemas tab in the Navigator, you can see all of your databases.

Selecting a Database

To select a database in MySQL Workbench, simple double-click it in the Schemas tab. The database is expanded, and its name becomes bold to indicate that it's ready for use.

Note

It's Using `USE` Earlier you learned about the `USE` statement, which opens a database. When you double-click on a database in MySQL Workbench, the tool is using a `USE` statement for you automatically. In fact, every option in MySQL Workbench uses SQL statements for you.

Learning About Databases and Tables

To see details about databases and tables, simply click on them. Details are displayed in the Information panel beneath the Navigation panel, as shown in Figure 3.4.

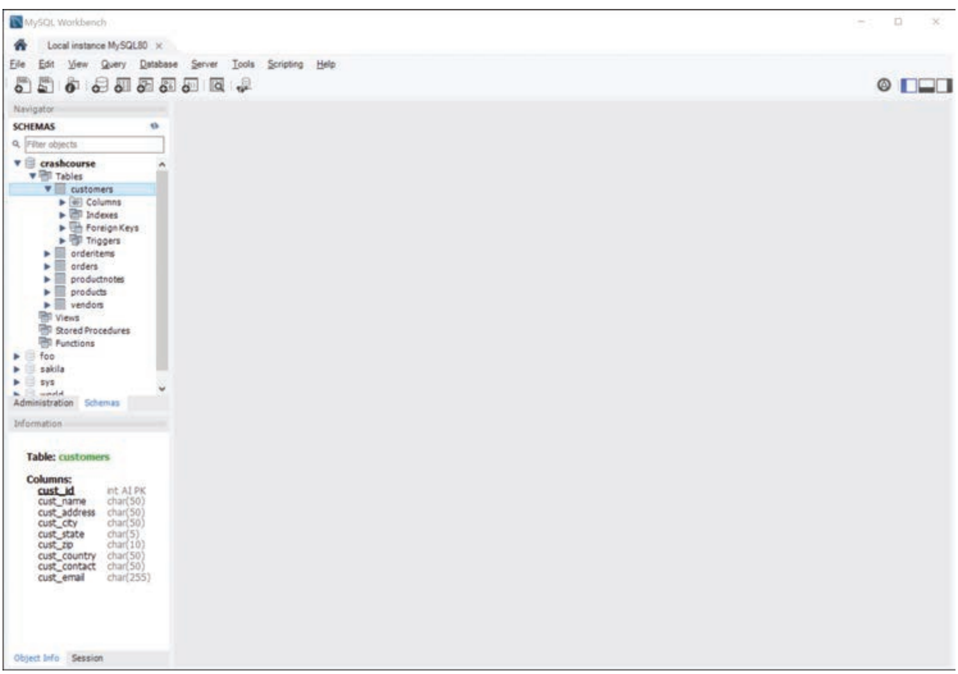

Figure 3.4 The Information panel below the Navigation panel.

Executing SQL Statements

Throughout this book, you'll be writing and testing SQL statements, and it's important to know how to do that with MySQL Workbench.

With a database open (any database will do), click the leftmost button on the toolbar; this is the New Query button, and it will open an editor window where you can type your SQL commands.

Figure 3.5 shows the database `world` being used, with the following SQL statement entered:

```
SELECT * FROM country;
```

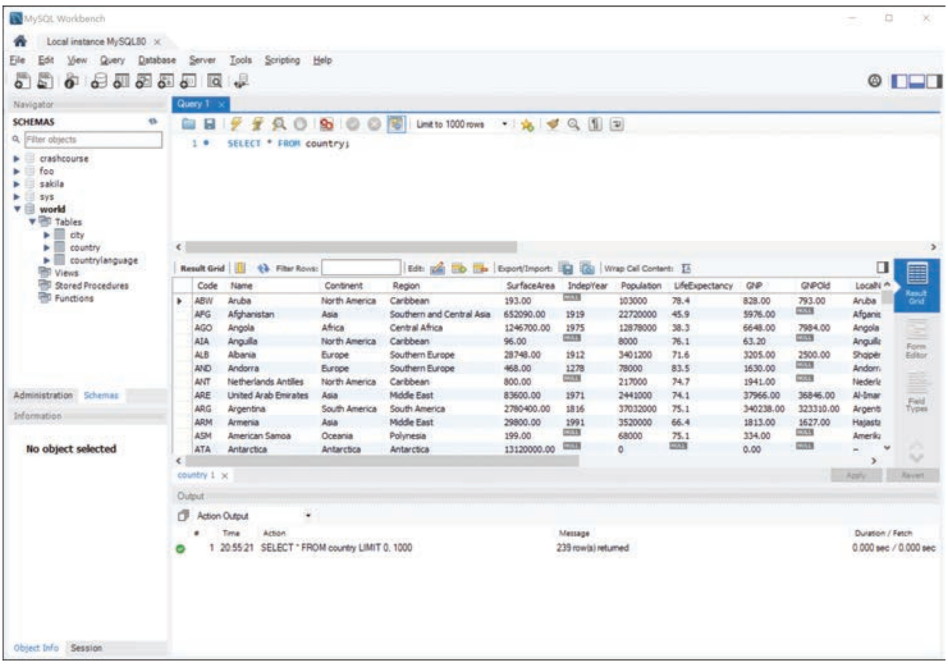

Figure 3.5 Query results are displayed in a grid below the SQL statement.

To execute (or run) the SQL statement, click the yellow lightning bolt button above the query window. The SQL statement is then executed, and results (if there are any) are displayed in a grid below.

Now you're ready to install the example database and tables and proceed with learning MySQL.

Next Steps

Now that you know how to connect and log in to your MySQL DBMS, you are ready to create the example database and tables used throughout this book.

Refer to Appendix B, "The Example Tables," for a detailed description of the tables as well as step-by-step instructions for installing them.

Summary

In this chapter, you learned how to connect and log in to MySQL and how to execute SQL commands by using both the command-line utility and MySQL Workbench. Armed with this knowledge, you can now dig into the all-important `SELECT` statement.